

WORK BREAKDOWN STRUCTURE (WBS)

Device Risk Analysis | Case Study No. 86

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1. Hierarchical Work Breakdown Structure

Code	Work Package	Deliverable Scope	Duration
1.0	Project Planning	Scope definition & requirements analysis	2 Days
2.0	Dataset Synthesis	3,012 telemetry records based on NIST benchmarks	3 Days
3.0	Data Cleaning	Deduplication & median/mode imputation	2 Days
4.0	Exploratory Analysis	Distributions, correlations, and feature comparisons	3 Days
5.0	Preprocessing	ColumnTransformer with StandardScaler & OneHotEncoder	3 Days
6.0	Model Benchmarking	6 ML algorithms trained under identical splits	4 Days
7.0	Model Tuning	5-fold Stratified GridSearchCV for optimal champion	3 Days
8.0	Streamlit Web App	6-page interactive dashboard with instant prediction	4 Days
9.0	Quality Assurance	Full test suite execution and verification	2 Days
10.0	Documentation	7 comprehensive DOCX and PDF documents	5 Days
11.0	Final Submission	Project report, viva presentation guide, and archive	2 Days

2. Critical Path & Quality Gates

The critical path spans Work Packages 2.0 -> 3.0 -> 5.0 -> 6.0 -> 7.0 -> 8.0 -> 10.0. Each milestone enforces strict validation criteria ensuring 100% reproducibility and flawless academic integrity.